

DAYTON INTL, OH, USA

WMO: 724290

Lat: 39.906N

Lon: 84.219W

Elev: 1003

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.7	7.5	-5.6	4.2	3.1	-0.7	5.5	8.6	29.0	30.2	26.9	30.5	9.4	270	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	90.1	73.7	87.9	72.8	85.5	71.4	76.6	86.0	75.2	83.8	73.8	81.9	10.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	129.9	81.9	72.3	124.3	80.3	71.1	119.0	78.8	40.6	85.9	39.2	84.1	37.9	81.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.0	19.4	-4.1	93.3	7.0	3.4	-9.2	95.7	-13.3	97.7	-17.3	99.6	-22.4	102.0		
(5)			-4.8	79.4	6.9	2.0	-9.7	80.8	-13.8	81.9	-17.6	83.0	-22.6	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	27.8	31.5	40.7	52.6	62.8	71.5	74.4	73.2	66.7	55.0	42.9	33.2		
	DBStd	18.82	12.65	11.69	11.32	9.47	8.43	6.20	5.25	5.11	7.27	8.96	9.70	10.67		
	HDD50	2477	694	526	335	80	7	0	0	0	0	50	255	530		
	HDD65	5442	1153	937	755	382	147	17	2	4	67	329	662	987		
	CDD50	3504	7	8	47	159	403	645	758	720	500	205	43	9		
	CDD65	994	0	0	2	12	79	212	294	259	116	20	0	0		
	CDH74	8085	0	0	15	91	610	1677	2498	2077	972	144	1	0		
	CDH80	2479	0	0	1	8	131	530	847	639	295	28	0	0		
	WSAvg	9.6	11.6	11.2	11.0	11.2	9.5	8.4	7.6	7.1	7.5	9.3	10.4	10.7		
	PrecAvg	38.6	2.6	2.2	3.4	3.9	4.1	4.0	3.7	3.1	3.0	2.6	3.1	3.0		
(15) Precipitation	PrecMax	59.8	9.2	5.8	7.6	8.7	9.0	9.5	8.5	8.0	10.8	6.3	8.1	10.0		
	PrecMin	24.2	0.3	0.2	1.1	0.6	1.5	0.3	0.5	0.0	0.3	0.2	0.7	0.6		
	PrecStd	7.7	1.5	1.2	1.5	1.8	2.0	1.7	1.6	1.8	2.1	1.5	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	67.0	75.8	81.0	86.7	91.6	94.4	91.7	91.3	83.9	70.7	63.5		
		MCWB	57.4	58.3	60.7	63.8	70.1	73.9	76.8	74.3	70.0	68.2	59.3	58.6		
	2%	DB	57.2	59.2	69.4	76.7	83.5	88.4	90.2	88.8	87.2	77.9	65.9	58.3		
		MCWB	54.5	52.7	57.5	61.4	68.0	72.5	75.1	73.6	70.1	64.7	57.2	54.2		
	5%	DB	52.0	54.1	63.8	72.8	80.8	85.7	87.5	86.3	83.2	73.6	62.4	53.4		
		MCWB	49.0	48.9	54.7	59.1	67.2	70.9	73.6	72.4	68.8	63.0	55.7	49.9		
	10%	DB	46.0	48.7	58.8	68.6	77.3	83.0	84.8	83.6	79.8	69.9	58.1	48.8		
		MCWB	42.3	43.3	51.4	57.1	65.5	69.8	72.2	71.0	66.7	61.3	52.0	45.1		
	0.4%	WB	58.4	59.6	62.5	67.2	73.0	77.3	79.1	77.6	75.0	71.2	62.8	59.4		
		MCDB	60.8	65.3	72.5	77.1	82.0	86.8	89.8	87.2	84.5	80.2	67.3	62.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	54.5	54.4	59.1	63.8	70.9	75.1	77.2	75.8	72.9	68.2	59.5	55.0		
		MCDB	57.0	57.8	66.8	72.4	80.3	84.3	87.0	84.5	81.9	74.0	64.1	57.4		
	5%	WB	49.7	49.2	56.2	61.4	69.0	73.4	75.5	74.6	71.1	65.2	56.4	50.4		
		MCDB	52.0	54.1	62.2	68.9	77.3	82.6	84.2	82.6	79.3	71.8	61.1	52.9		
	10%	WB	42.4	43.5	52.2	59.0	67.1	71.8	74.1	73.1	69.3	62.1	52.8	45.4		
		MCDB	45.3	48.4	58.3	66.8	74.9	80.1	82.0	80.5	76.1	68.7	57.6	48.5		
	5% DB	MDBR	14.2	15.3	17.3	20.1	18.6	17.9	17.9	18.4	19.9	18.6	15.9	13.0		
		MCDBR	17.5	21.5	22.5	23.7	21.2	20.6	19.5	20.1	23.2	22.7	20.1	18.0		
		MCWBR	15.3	16.9	14.0	13.6	10.2	9.2	7.9	8.1	9.7	12.2	14.3	15.6		
		MCWBR	16.5	20.4	20.0	20.3	18.2	17.6	17.6	16.9	18.5	19.5	18.1	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.323	0.343	0.391	0.435	0.442	0.445	0.439	0.392	0.353	0.322	0.299		
		taud	2.441	2.351	2.373	2.253	2.174	2.202	2.204	2.203	2.312	2.418	2.462	2.464		
	Ebn at Noon		276	283	289	280	269	266	264	262	268	269	265	267		
		Edh at Noon	25	32	35	42	47	45	45	44	37	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		538	814	1131	1520	1741	1926	1890	1733	1414	991	663	454		
	RadStd		48	94	97	128	145	130	110	85	129	91	80	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+108
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+117

Nomenclature: See separate page